

```
$ javac NotificationLab.java && java NotificationLab
```

# Java Interface

// □□□□□□□□□□

## contract

5/5

## dispatch

5/5/2019

## injection

5/5/2019

```
$ cat learning-path.md
```



01 / NOT COMPLETED

interface NOT COMPLETED implements

02 / NOT COMPLETED

□□□□□□□□

03 / NOT COMPLETED

constructor injection□DIP□ISP

04 / NOT COMPLETED

default□lambda□sealed

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## PART 01



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```
1  class NotificationService {
2      void notify(String channel, String message) {
3          if (channel.equals("email")) {
4              new EmailClient().deliver(message);
5          } else if (channel.equals("sms")) {
6              new SmsClient().deliver(message);
7          }
8      }
9  }
```

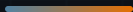
#### COUPLING REPORT

× □□□□□□□

× □□ SDK □□□□

× □□□□ SDK □□□□

□□ Push → □□□□□□

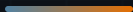


DOMAIN LANGUAGE

Notifier

send(message)





```
1 public interface Notifier {  
2     void send(String message);  
3 }
```

**public abstract**

□□□□□□□□□□

**no instance state**

□□□□□□□□□□□□

**role name**

□□□□□□□□□□

**contract** ≠ □□□□□□□□□□□□□□□□

# implements □□□□□

```
1  final class EmailNotifier implements Notifier {
2      @Override
3      public void send(String message) {
4          System.out.println("EMAIL > " + message);
5      }
6  }
7
8  final class SmsNotifier implements Notifier {
9      @Override
10     public void send(String message) {
11         System.out.println("SMS > " + message);
12     }
13 }
```

□□□□□□□□ public □□□□□□□□□□□□□□□□□ package-private□

## PART 02



□□□□□□□□□□





```
1 Notifier notifier = new EmailNotifier();
2 notifier.send("Welcome");
3
4 notifier = new SmsNotifier();
5 notifier.send("Code: 8042");
```

COMPILE TIME

Notifier 000 send 0

RUN TIME

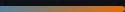
0000000000

00000000 → 00000000



```
1 final class NotificationService {  
2     private final Notifier notifier;  
3  
4     NotificationService(Notifier notifier) {  
5         this.notifier = notifier;  
6     }  
7  
8     void notify(String message) {  
9         notifier.send(message);  
10    }  
11 }
```

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```
1  final class NotificationService {  
2      private final Notifier notifier;  
3  
4      NotificationService(Notifier notifier) {  
5          this.notifier = notifier;  
6      }  
7  }
```

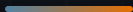
```
final NotificationService
```



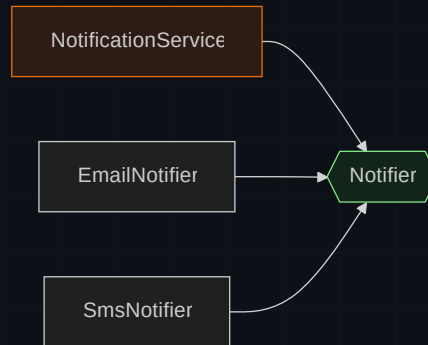
```
1  public static void main(String[] args) {  
2      Notifier notifier =  
3          new EmailNotifier();  
4  
5      var service =  
6          new NotificationService(notifier);  
7  }
```

COMPOSITION ROOT

NotificationService



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□□□□□□□□□□



xxxxxxxxxxxxxxxxxxxxxxxx

xxxx

xxxx

Email

SMS

Push

```
$ deps --graph
```

```
Service → Notifier ← implementations
```

```
xxxx → xx Notifierxxxxxx
```

▶ dispatch()

```
// select channels, then dispatch
```

## PART 03



DIP[ISP] [ ][ ][ ][ ][ ][ ]

# interface ≠ DIP

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CHECKPOINT\_01

00000000 DIP0

A 0000 interface000000 DIP

B 000000000000000000000000

C 00 class 00000000 interface



```
1 interface NotificationPlatform {  
2     void send(String message);  
3     void schedule(String message, Instant at);  
4     void revoke(String id);  
5 }
```

SMS `send` `schedule` `revoke` `UnsupportedOperationException` □□□□□□□□□□

Notifier

Schedulable

Revocable



```
1  final class CompositeNotifier implements Notifier {
2      private final List<Notifier> delegates;
3
4      CompositeNotifier(List<Notifier> delegates) {
5          this.delegates = List.copyOf(delegates);
6      }
7
8      public void send(String message) {
9          for (Notifier notifier : delegates) {
10             notifier.send(message);
11         }
12     }
13 }
```

Notifier



```
1  final class RecordingNotifier
2      implements Notifier {
3
4      final List<String> messages =
5          new ArrayList<>();
6
7      public void send(String message) {
8          messages.add(message);
9      }
10 }
```



```
1  @Test
2  void sendMessage() {
3      var fake = new RecordingNotifier();
4      var service =
5          new NotificationService(fake);
6
7      service.notify("shipped");
8
9      assertEquals(
10         List.of("shipped"),
11         fake.messages);
12 }
```

fast · deterministic · offline

# Interface abstract class

|  | Interface                                                                                                                                                                   | Abstract class                                                                      |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |                                                                                            |  |
|  |                                                                                            |  |
|  |                                                                                            |  |
|  |                                                                                            |  |
|  |   default |  |



## PART 04



Java 8 9017 00000

# default

NOTES

# static

Java8+

```
1 interface Notifier {
2     void send(String message);
3
4     default void sendUrgent(String message) {
5         send("[URGENT] " + message);
6     }
7
8     static Notifier silent() {
9         return message -> { };
10    }
11 }
```

## default

API

## static

# private interface method Java 9+

```
1  interface Notifier {
2      default void sendUrgent(String message) {
3          send("[URGENT] " + normalize(message));
4      }
5
6      default void sendNormal(String message) {
7          send(normalize(message));
8      }
9
10     private String normalize(String message) {
11         return message.strip();
12     }
13
14     void send(String message);
15 }
```

private □□□□□□□□□□□□□□□□□□

# Functional Interface + lambda

Java8+

```
1  @FunctionalInterface
2  interface Notifier {
3      void send(String message);
4  }
5
6  Notifier console =
7      message -> System.out.println(message);
```

□□□□□□

default / static / private □□□

**@FunctionalInterface**

□□□□□□□□□□



# Sealed interface Java 17

```
1 sealed interface DeliveryResult
2     permits Delivered, Rejected, Retrying {}
3
4 record Delivered(String id) implements DeliveryResult {}
5 record Rejected(String reason) implements DeliveryResult {}
6 record Retrying(int attempt) implements DeliveryResult {}
```

plugin API

Java 15 ☒ ☒ ☒ ☒ JEP 409 ☒ Java 17 ☐ ☐ ☐

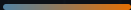
# Monaco

Java

```
interface Notifier {
    // TODO: send(String message)
}

final class PushNotifier implements Notifier {
    // TODO: "PUSH > " + message
}

final class NotificationService {
    // TODO: Notifier
}
```



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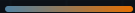
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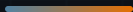
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fake



| □□                            | □□      | □□       |
|-------------------------------|---------|----------|
| default / static methods      | Java 8  | □□□□□□□□ |
| Functional interface / lambda | Java 8  | □□□□□□□□ |
| private interface methods     | Java 9  | □□□□□□   |
| Sealed interfaces             | Java 17 | □□□□□□□□ |

□□□□JLS §9 · JLS §9.8 · JEP 409



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fake NOT DESIGN □□□□□

Interface □□□□□□□□□□□□□□

# Lab complete.

~~~~~ FailingNotifier ~~~~~

```
$ java NotificationLab --reflect
```

```
contract → dispatch → injection → test
```